EMS622U-SUSTAINABILITY ASSESSMENTS FOR DESIGN

FULL LIFE CYCLE ANALYSIS OF TAMPONS WITH PLASTIC APPLICATORS VS. SILICONE MENSTRUAL CUPS

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1 INTRODUCTION

1.1 Products being studied

- Single-use tampons with plastic applicators
- Reusable silicone menstrual cups

1.1.1 Importance of the Study

Menstrual products are essential for over half the global population. With around 1.8 billion people menstruating monthly (UNICEF, 2019) and over 200,000 tonnes of menstrual waste being produced in the UK alone each year, the sustainability of menstrual hygiene products is a significant and under-addressed challenge which highlights the need for this study (Shout Out UK, 2024). Despite an increase in sustainability-focused initiatives for consumer goods, the menstrual product market remains dominated by single-use products such as tampons and pads (UNEP, 2022), with studies showing that over 40% of menstruators in developed countries prefer tampons as their primary product (Future Market insights, 2025).

The continued reliance on single-use tampons, especially those with plastic applicators, contributes to significant environmental impact such as greenhouse gas emissions, marine microplastic pollution, and landfill waste (Fourcassier et al., 2022). This makes the study of menstrual hygiene products a great opportunity to reduce environmental impact if alternative products are used or conventional products are improved. There are also many emerging concerns about health risks associated with ingredients used in tampons such as dioxins and synthetic fragrances, as well as economic concerns including period poverty—the inability to afford appropriate menstrual protection, and the long-term affordability of tampons (Hait & Powers, 2019).

Reusable silicone menstrual cups on the other hand are a more sustainable alternative with significantly lower environmental impact and lifetime cost (Hait & Powers, 2019). However, barriers including lack of public education, stigma, and cleaning infrastructure continue to limit their adoption (Davidson, 2012).

1.2 Objectives of the Study

The objectives being followed to carry out this study are to:

1. Conduct an environmental, economic and social Life Cycle Assessment (LCA) of the two products using the following sustainability indicators:

Environmental:

- Global Warming Potential (GWP).
- Human Toxicity Potential.

Economic:

- Cost per use

Social:

- Consumer health and safety risks

2. Identify **lifecycle hotspots** contributing to the most environmental, economic and social impacts.

3. Conduct a **sensitivity analysis** of the data and assumptions adopted and their impact on the results.

5. Recommend **design and policy changes** to improve the sustainability of menstrual products and adoption of more sustainable alternatives.

2 ENVIRONMENTAL ASSESSMENTS

2.1 Product Selection and Assessment Goals

This study examined two menstrual hygiene products for comparison: single-use tampons with plastic applicators, representing a mainstream disposable option widely used in the UK, and the reusable silicone menstrual cup, representing a more sustainable alternative (Hait & Powers, 2019).

Popular products produced by major brands in the UK market were selected as representative examples. According to 2021 data, Tampax Compak *(Figure 1)* was the leading tampon brand in Great Britain, with approximately 2.16 million users (Statista, 2021). As primary materials are consistent across manufacturers, the tampon model assessed reflects the average UK-market tampon. The menstrual cup market shows similar consistency in composition, typically relying on medical-grade silicone. The DivaCup *(Figure 2)* was used as the representative menstrual cup due to its popularity.



Figure 1: Tampon with plastic applicator (Tampax.co.uk, 2021).

Figure 2: Reusable Silicone Menstrual Cup (DIVA Canada, 2025).

Goal:

The goal of the environmental LCA was to estimate the Global Warming and Human Toxicity Potentials of single-use tampons with plastic applicators as compared to reusable silicone menstrual cups, identify life cycle hotspots and emission reduction opportunities for future improvements in the menstrual hygiene market, and provide information on how menstruators can reduce the environmental impacts linked to the use of menstrual products.

2.2 System Definition: Boundaries and Scoping

This study modelled the cradle-to-grave LCA *(Figure 3)* for both products over a ten-year usage period in the United Kingdom. This is an evaluation of the mass and energy inputs as well as the pollution and waste of both products across all stages of their life cycles specified below:

1. Raw material extraction and processing

2. Manufacturing and assembly

3. Transportation

4. End use

5. End-of-life disposal

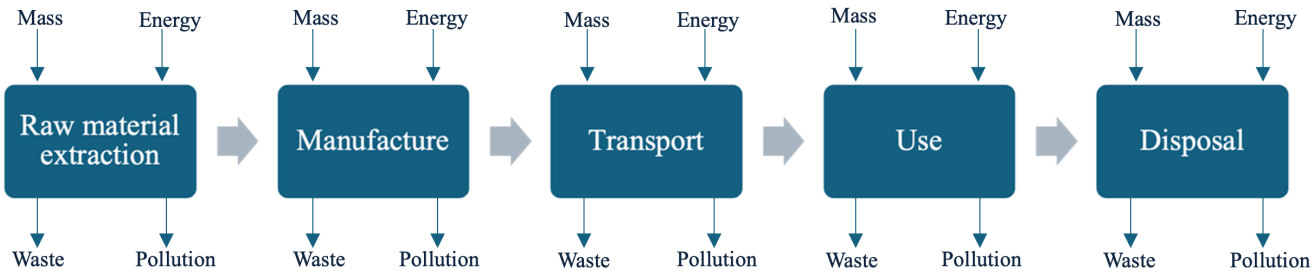


Figure 3: Cradle-to-grave life cycle stages

2.2.1 Functional Unit

One tampon typically provides up to 8 hours of menstrual protection, equating to 4 tampons a day. Most estimates thus suggest an average of 240 tampons per person per year assuming average 5-day periods among menstruators. One reusable menstrual cup however lasts up to ten years and emptied at least every twelve hours during menstruation (Hait & Powers, 2019). To enable a fair comparison in this study, the functional unit was thus defined as:

"The number of products required to provide effective menstrual protection for one person over ten years."

This equates to **2400 tampons** (240 per year) versus **1 menstrual cup** (lifespan = 10 years).

2.2.2 Impact Categories

This environmental LCA evaluated two environmental midpoint indicators:

2. **Global Warming Potential (GWP)** (gCO2eq.) **:** measures the ability of a greenhouse gas to trap heat in the atmosphere, contributing to climate change relative to carbon dioxide. GWP is expressed in grams of carbon dioxide equivalent (g CO_2-eq). This metric is standardised under the Intergovernmental Panel on Climate Change (IPCC) (Bogner et al., 2008).
3. **Human Toxicity Potential** (gDCBeq.) **:** reflects the potential harm to human health that certain substances can cause across a product's life cycle through ingestion, inhalation, or dermal exposure. It is expressed in kilograms of 1,4-dichlorobenzene (1,4-DB) equivalents, a reference compound used to standardise toxicity impacts (Durante et al., 2015).

These indicators reflect the primary environmental burdens associated with menstrual products across their respective lifecycles (Hochwalt et al., 2023).

2.3 Inventory Analysis, Assumptions and Justifications

This section outlines the inventory data sources relied upon for assumptions used in the study. This includes all the input and output data across all life cycle stages for both products, normalised to the functional unit : **2400 tampons** or **1 menstrual cup**.

A collection of research papers, government reports, large databases, manufacturer reports and private sector reports were used to collect inventory data, and to guide assumptions. Where UK-specific data was limited, assumptions were developed based on European averages.

2.3.1 Raw materials

The material composition of both products was identified based on manufacturer specifications. Both Tampax and DivaCup list the materials in each of their products on their respective websites. The menstrual cup material list was straight forward, as the cup is composed only of medical-grade silicone like most other menstrual cups in the market, however, tampons are each composed of several components (Hait & Powers, 2019). *Tables 1-2* show the average total weight and composition of tampons and cups measured by Hait & Powers as manufacturers provided no data on the composition by mass of the products.

Table 1 Mass, percentage composition, and materials used in each component of a typical UK-market tampon.

Component (Tampax, 2025)	**Mass (g) (Hait & Powers, 2019)**	**% of Total Mass (Hait & Powers, 2019)**	**Material (Tampax, 2025)**
Absorbent Fiber	1.4290	27.8%	Rayon
Removal String	0.1000	1.9%	Polyester (50%), Cotton (50%)
Nonwoven Cover (Overwrap)	0.0989	1.9%	Polyester, Rayon
Applicator	2.6386	51.4%	Polyethylene (LDPE, injection moulded)
Total	**5.137 g**	**100%**	—

Table 2: total mass and material of a typical silicone menstrual cup.

Component (DivaCup, 2025)	**Mass (g) (Hait & Powers, 2019)**	**% of Total Mass (Hait & Powers, 2019)**	**Material (Hait & Powers, 2019)**
Silicone Cup	14.8	100%	Medical-grade silicone
Total	**14.8 g**	**100%**	—

2.3.2 Transport Distances and Modes (Faraca et al., 2023)

Transport distance and mode assumptions were informed by the European Commission Joint Research Centre's revision report on EU Ecolabel criteria, which used the Product Environmental Footprint (PEF) methodology to evaluate lifecycle impacts of absorbent hygiene products and reusable menstrual cups. (Faraca et al., 2023). *Tables 3-4* summarise the transport modes, distances, and justifications for the assumptions used to model the transport stage of both products.

Table 3: Tampon transport modes, distances, and justifications ((Faraca et al., 2023)

Stage	Transport Type	Distance (km)	Justification
Raw materials → Manufacturer	Sea, container ship	~10,000	• Imported materials like cotton and rayon are often sourced from North America or Asia. • Distance reflects how raw materials are typically imported by container ship and then transported to manufacturers by truck within Europe.
Manufacturer → Retailer	Truck >32t, EURO 4	~3500	• Tampons are typically manufactured in Europe and transported to UK retailers via truck. • Transport distance reflects typical travel distances from the factories to retailers within Europe.
Retailer → End User	Light commercial vehicle	~5	• Typical travel distance for consumer round trips to high-street stores based on European travel survey data.
End User → Landfill	Truck 16–32t, EURO 6	~100	• Used tampons are typically disposed of in household waste, which is collected by refuse trucks and transported to landfill facilities. • The transport distance is assumed to be ~100km based on European waste management data.

Table 4:Cup transport modes, distances, and justifications

Stage	Transport Type	Distance (km)	Justification
Raw materials → Manufacturer	Truck >32t, EURO 4	~1000	• Raw silicone is often sourced from European suppliers and typically shipped by lorry from chemical suppliers to European manufacturers.
Manufacturer → Retailer	Truck >32t, EURO 4	~3500	• Like tampons, finished cups are transported from European manufacturers to UK retailers via truck. • Transport distances are thus assumed to be the same for both products typical Europe-to-UK shipping distances.
Retailer → End User	Light commercial vehicle	~5	• Typical travel distance for consumer round trips to high-street stores based on European travel survey data. • Travel distances are assumed to be the same for both products.
End User → Landfill	Truck 16–32t, EURO 6	~100	• Like tampons, menstrual cups are disposed of in household waste collected by refuse trucks and transported to landfill facilities. • Transport distances are assumed to be the same for both products.

2.3.3 Use Phase

Assumptions related to menstrual cup sterilisation and user hygiene routines were adopted the comparative LCA conducted by GreenDelta (Vilabrille Paz et al., 2020), a study commissioned by Zero Waste Europe and conducted by a leading independent LCA consultancy. The report adheres to ISO 14040/14044 standards and applies established mid-point impact methods, making it a credible source for modelling realistic use-phase behaviours.

- **Tampons**: Are not reusable and so disposed of after use. No energy or water use assumed during the use phase.
- **Cups**: Typically sterilised once a month. Use phase thus considers sterilisation of the cup before the first use and between cycles by boiling using a kettle, but not washing of the cup due to variable hand washing habits applied during the use of both tampons and menstrual cups.

Table 5 outlines the energy usage calculations for monthly sterilisation of the menstrual cup via boiling in an electric kettle.

Table 5: Menstrual Cup Use-Phase Energy Assumptions (Vilabrille Paz et al., 2020)

Parameter	Value	Notes
Boiling duration	5 minutes	Once per month
Frequency	12 times per year	One sterilisation per menstrual cycle
Total annual boiling time	60 minutes (1 hour)	5 min × 12 months
Appliance power rating	2.5 kW	Average UK electric kettle
Annual energy consumption	2.5 kWh (≈9 MJ)	2.5 kW × 1 hour
Average daily use (equiv.)	0.0027 hours/day (10 seconds)	For modelling daily average input if needed

2.3.4 End-of-Life Scenarios

User behaviour was modelled based on usual theoretical assumptions. Scenarios such as flushing products down the toilet, although known to occur, were not accounted for in this study. However, given their potentially severe environmental impacts, particularly for plastic-containing products like tampons, such behaviours may influence outcomes and are considered in the sensitivity analysis (Vilabrille Paz et al., 2020).

- Tampons: Assumed to be 100% landfilled.
- Menstrual Cup: Assumed to be 100% landfilled due to lack of scalable silicone recycling in the UK analysis (Vilabrille Paz et al., 2020).

2.3.5 Exclusions and Cut-off points

- Adhesives and printing were excluded due to their marginal contribution (<1% total mass), following ISO 14044 cut-off rules.
- Packaging was excluded to ensure a fair and representative comparison between product types. Tampon and cup packaging varies significantly across brands—from plastic wraps to recyclable

cardboard—making it a brand-dependent factor. Including it would introduce variability not intrinsic to the product's core function. Furthermore, packaging constitutes a small percentage of the total mass and environmental burden, and would not materially change the comparative outcome, according to ISO 14044 cut-off rules.

- Additional handwashing before inserting or removing menstrual products was excluded from the analysis, as it is assumed to occur equally for both product types and varies significantly based on individual user habits.

2.4 Environmental Impact Assessment

This section presents the comparative environmental impact of 2400 single-use tampons with plastic applicators and one reusable menstrual cup; both modelled over a 10-year life cycle using two midpoint impact categories: **Global Warming Potential (GWP)** and **Human Toxicity Potential (HTP)**. All impact values are based on outputs from the Granta EduPack Eco Audit Tool and/or literature-verified data.

2.4.1 Global Warming Potential (GWP) Results

Summary of GWP Results

Over the 10-year functional unit, the total GWP of tampons was ***46.91 kg CO_2-eq***, whereas the reusable menstrual cup emitted just ***0.34 kg CO_2-eq***. *Table 6* presents the total carbon footprint of 2400 tampons and one menstrual cup over ten years, divided across material, manufacturing, transport, use, and disposal phases.

Table 6:Total GWP by Life Cycle Phase (kg CO_2-eq) for both products

Life Cycle Phase	Tampons (kg CO_2-eq)	Menstrual Cup (kg CO_2-eq)
Material	31.89	0.11
Manufacture	9.71	0.02
Transport	5.20	0.01
Use	0.00	0.21
Disposal	0.14	0.00
Total	**46.91**	**0.34**

2.4.2 Tampons Detailed Results

Table 7 shows the percentage of total GWP contributed by each life cycle phase for tampons, and *Figure 4* is a pie chart illustrating how GWP is distributed across material, manufacturing, transport, use, and disposal stages for tampons.

Table 7: Tampons –GWP Breakdown by life cycle stage

Phase	CO_2-eq (kg)	% of Total Impact
Material	31.89	68.00%
Manufacture	9.71	20.70%
Transport	5.17	11.00%
Use	0	0.00%
Disposal	0.14	0.30%
Total	**46.91**	**100%**

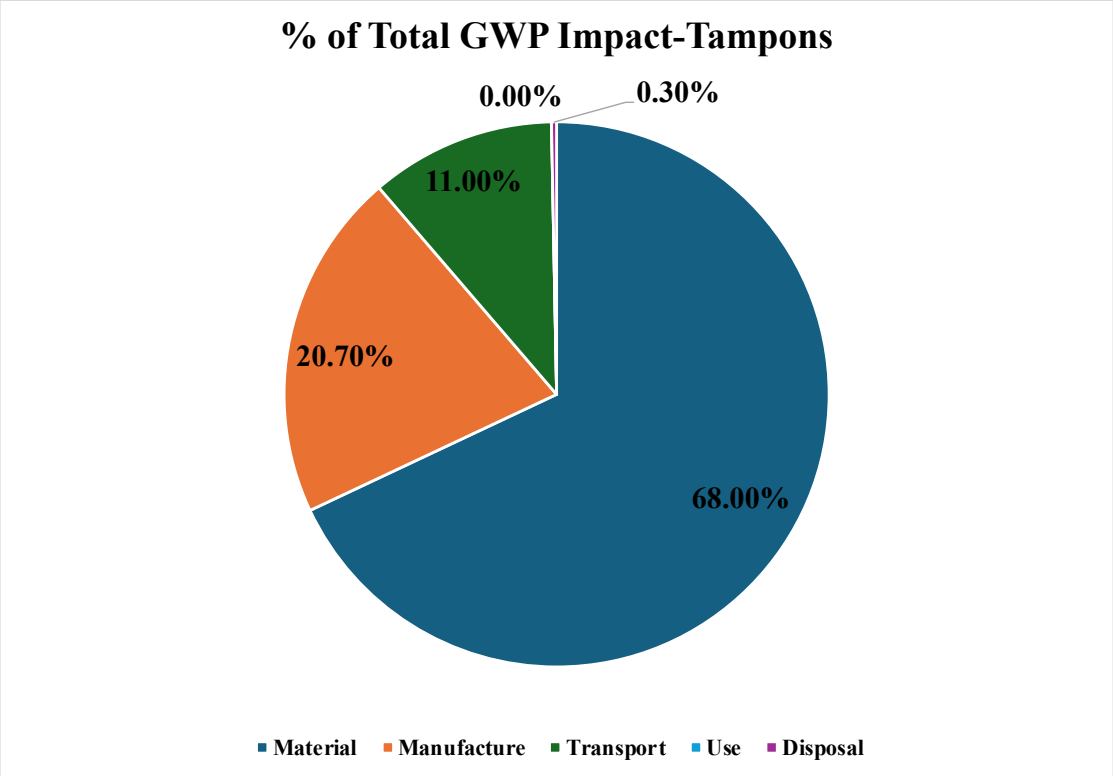


Figure 4: % of Total GWP Impact – Tampons

2.4.3 Tampons–Detailed Breakdown by Phase

Material Phase

Figure 5 below shows the proportion of GWP from individual components during the material production phase of tampons.

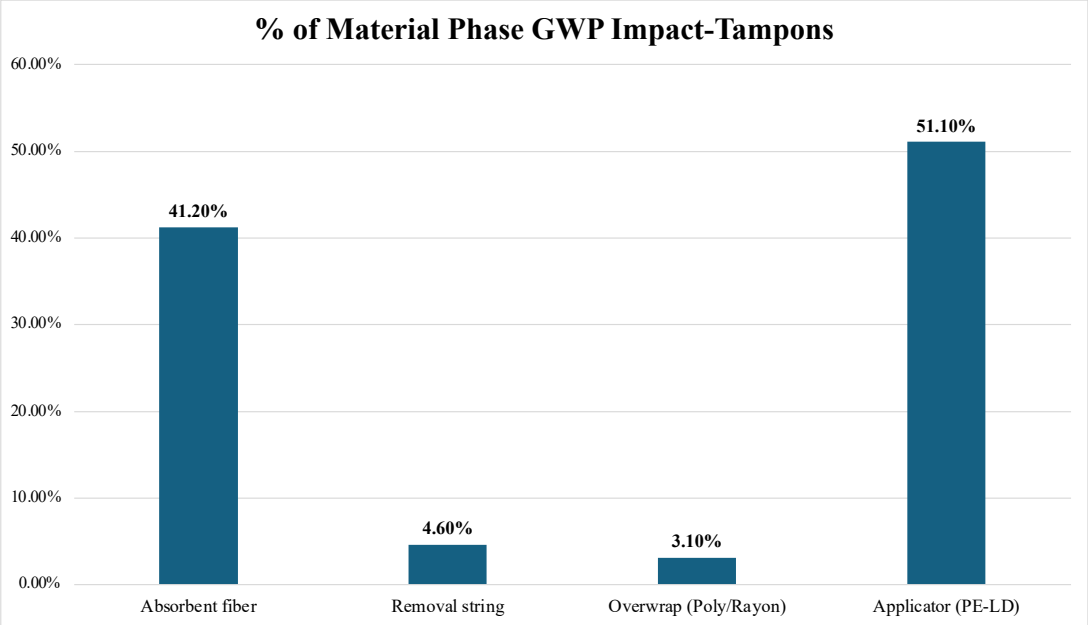


Figure 5: % of Material Phase GWP – Tampons

Manufacturing

Figure 6 below shows the breakdown of manufacturing-related GWP by each tampon component.

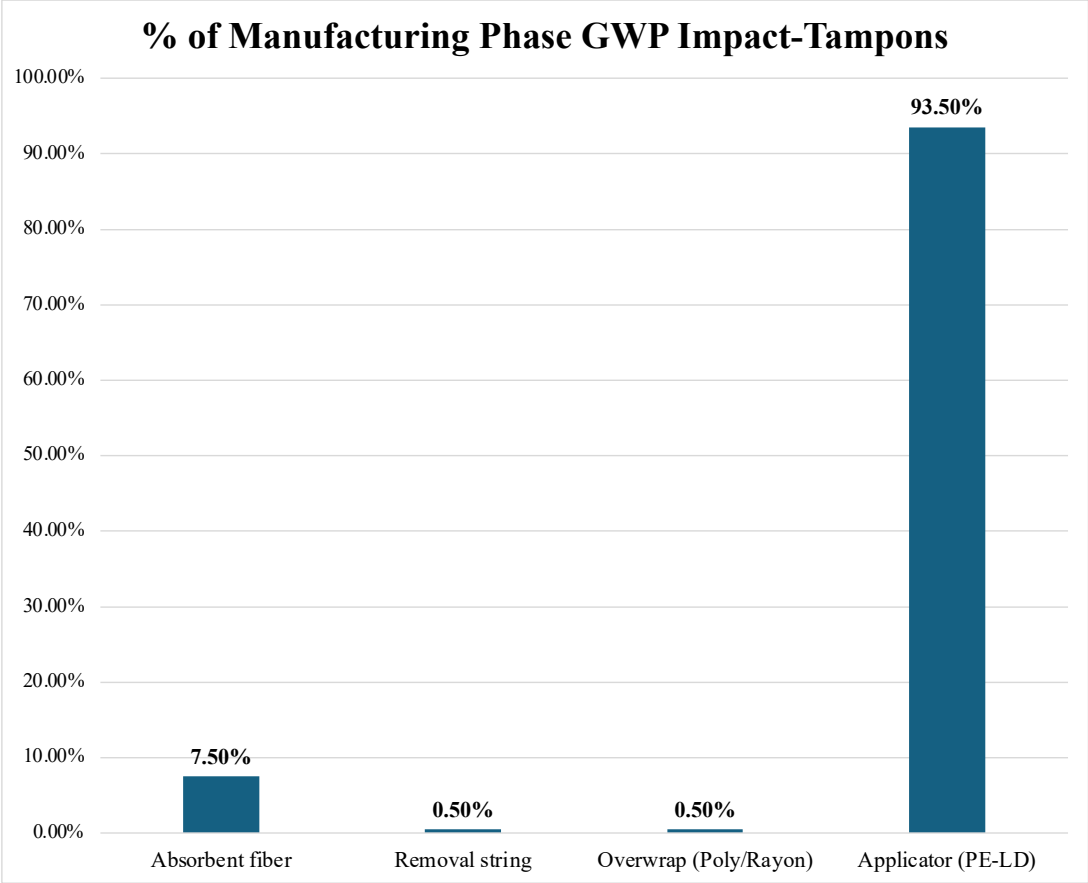


Figure 6: % of Manufacturing Phase GWP – Tampons

Transport

Figure 7 below presents the share of GWP from each transport stage in the tampon lifecycle, followed by *Figure 8* presenting the percentage GWP impact by component in this stage.

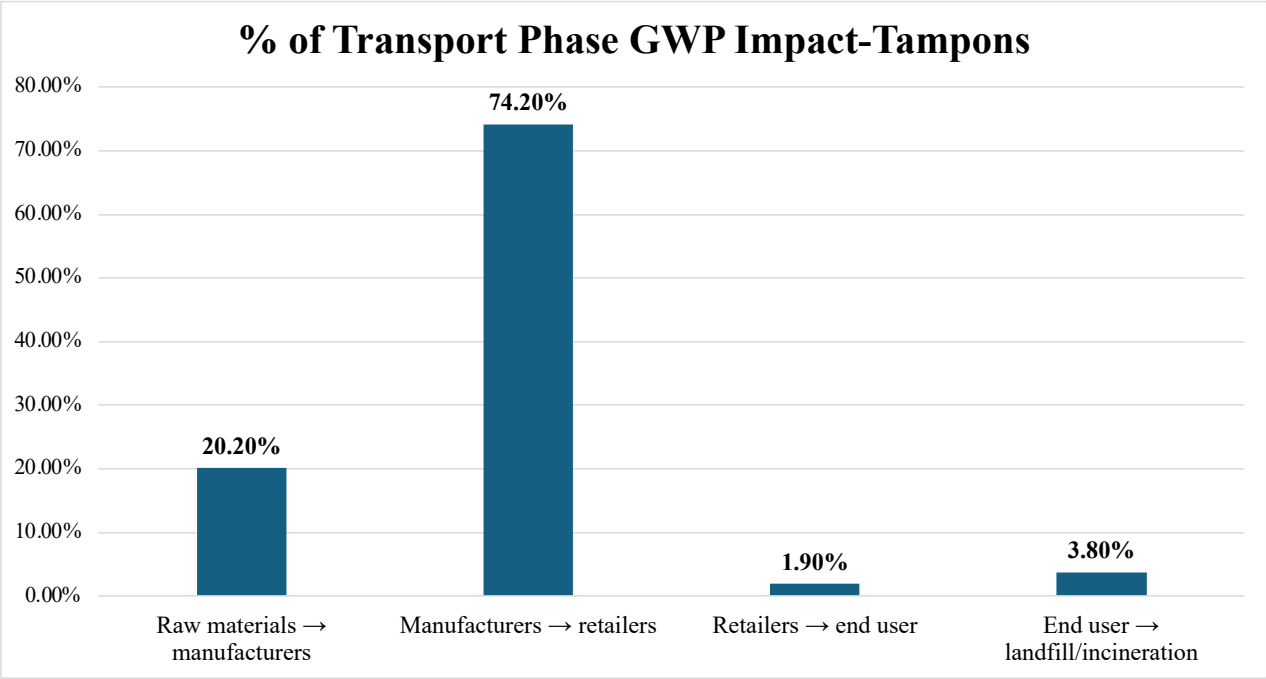


Figure 7: % of Transport Phase GWP – Tampons

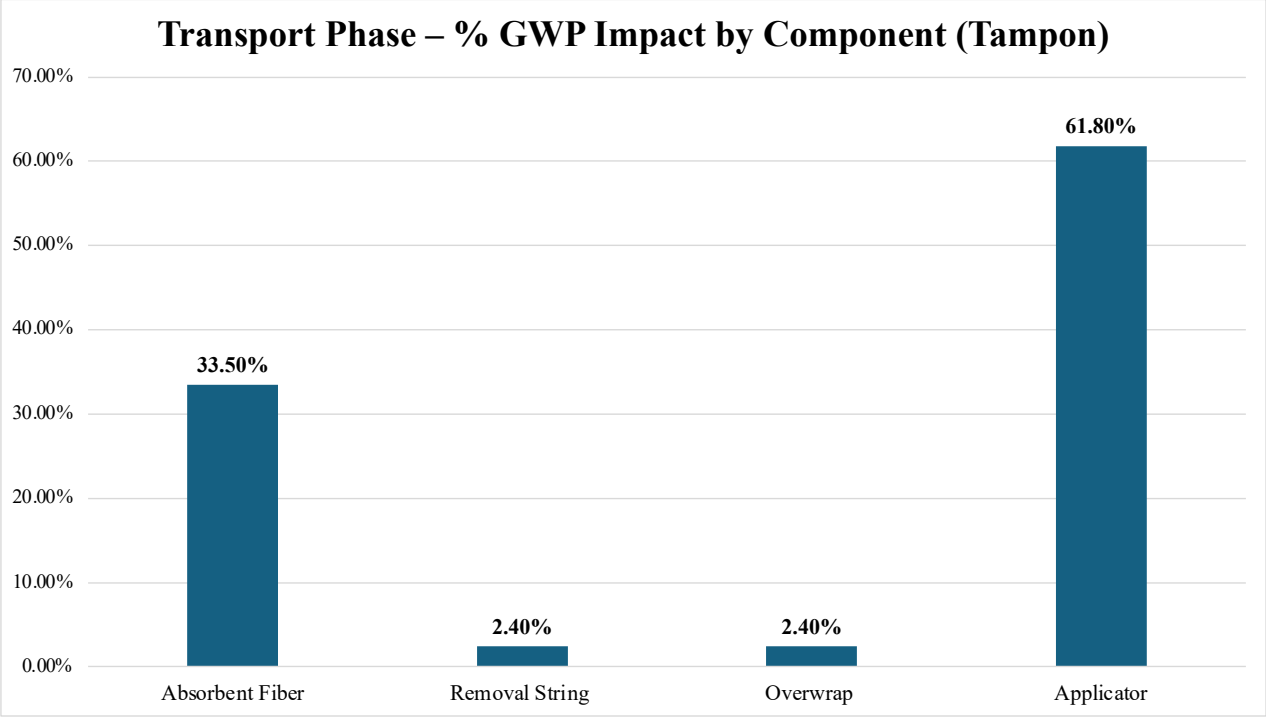


Figure 8: % of Transport Phase GWP by component – Tampons

Disposal

Figure 9 below shows the GWP contribution of different components during disposal of tampons.

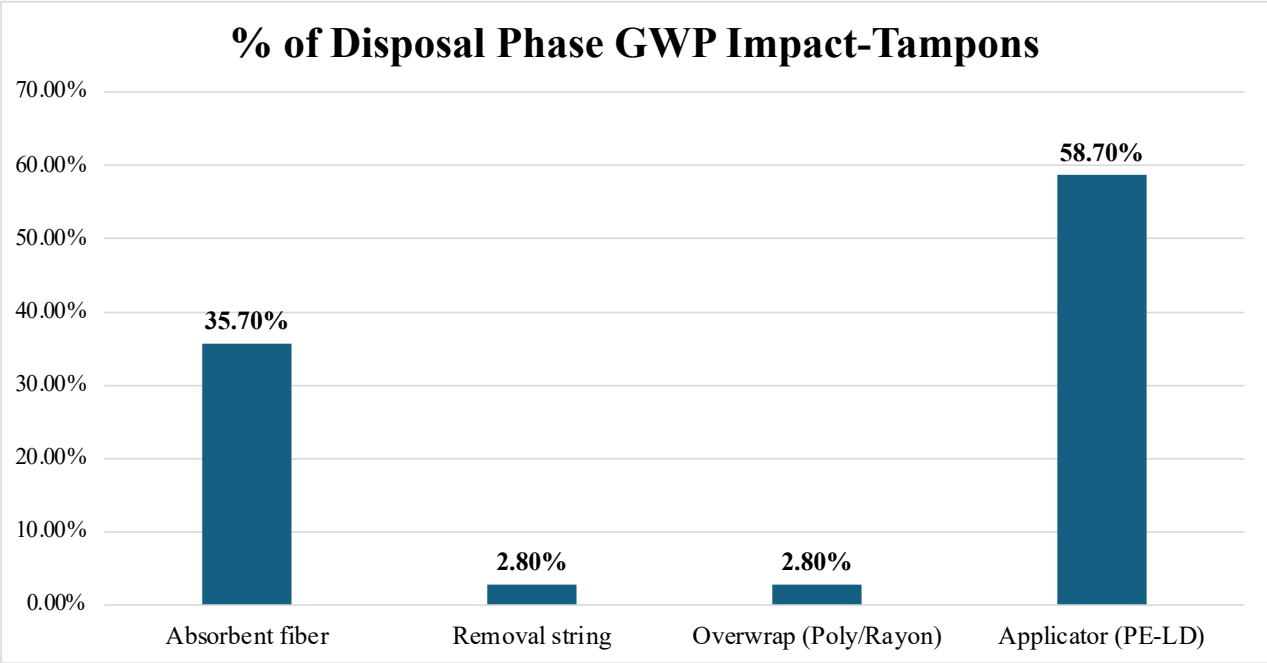


Figure 9: % of Disposal Phase GWP – Tampons

2.4.3 Cups Detailed results

Table 8 presents the kg CO_2-eq and percentage share for each life cycle phase of the menstrual cup and *Figure 10* is a pie chart displaying the share of GWP from each phase of the menstrual cup's lifecycle.

As menstrual cups are made of only one material (medical-grade silicone), all impacts are 100% silicone related throughout the life cycle stages.

Table 8: Menstrual Cup – GWP Breakdown by Phase

Phase	CO_2-eq (kg)	% of Total Impact
Material	0.1075	31.30%
Manufacture	0.0175	5.10%
Transport	0.0075	2.20%
Use	0.2109	61.30%
Disposal	0.0002	0.10%
Total	**0.3438**	**100%**

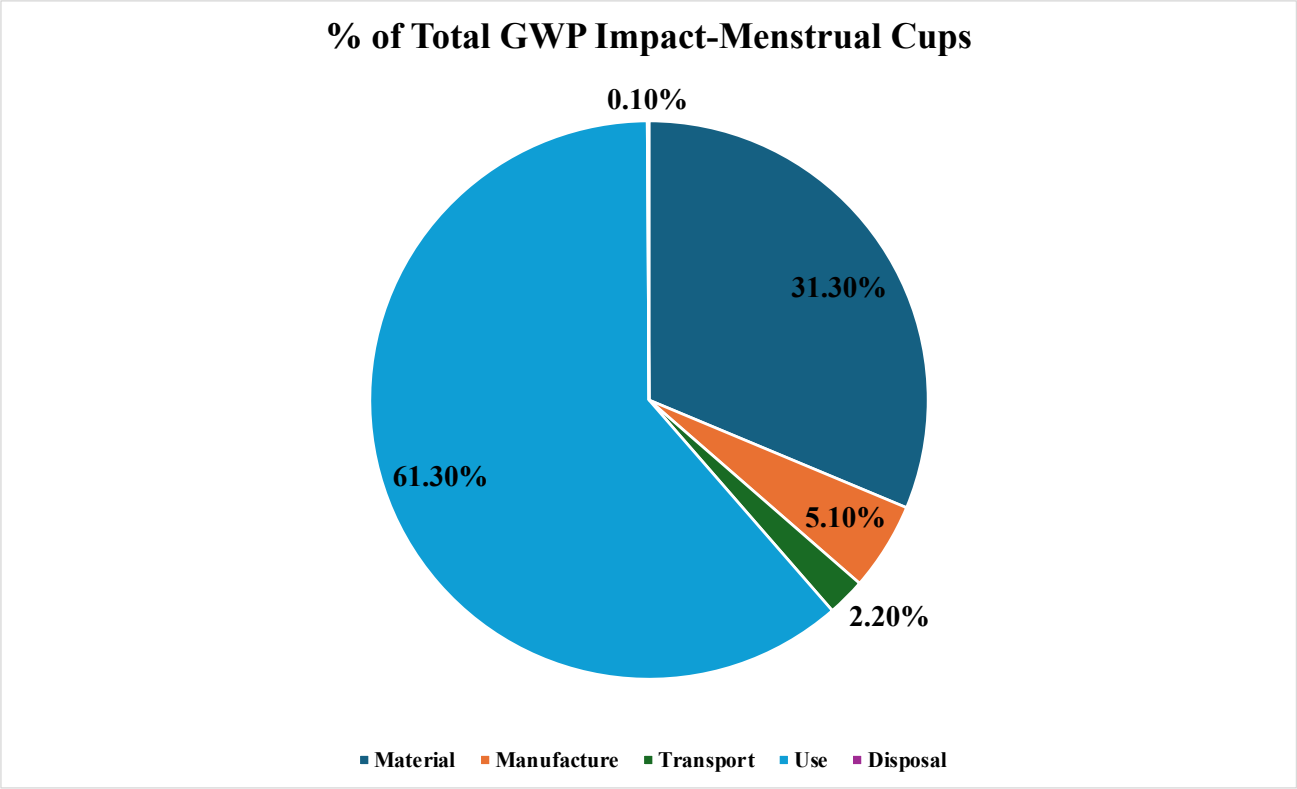


Figure 10: Pie chart showing how emissions are distributed across the menstrual cup's life cycle stages.

Transport

Figure 11 below breaks down the cup's GWP across its transport stages.

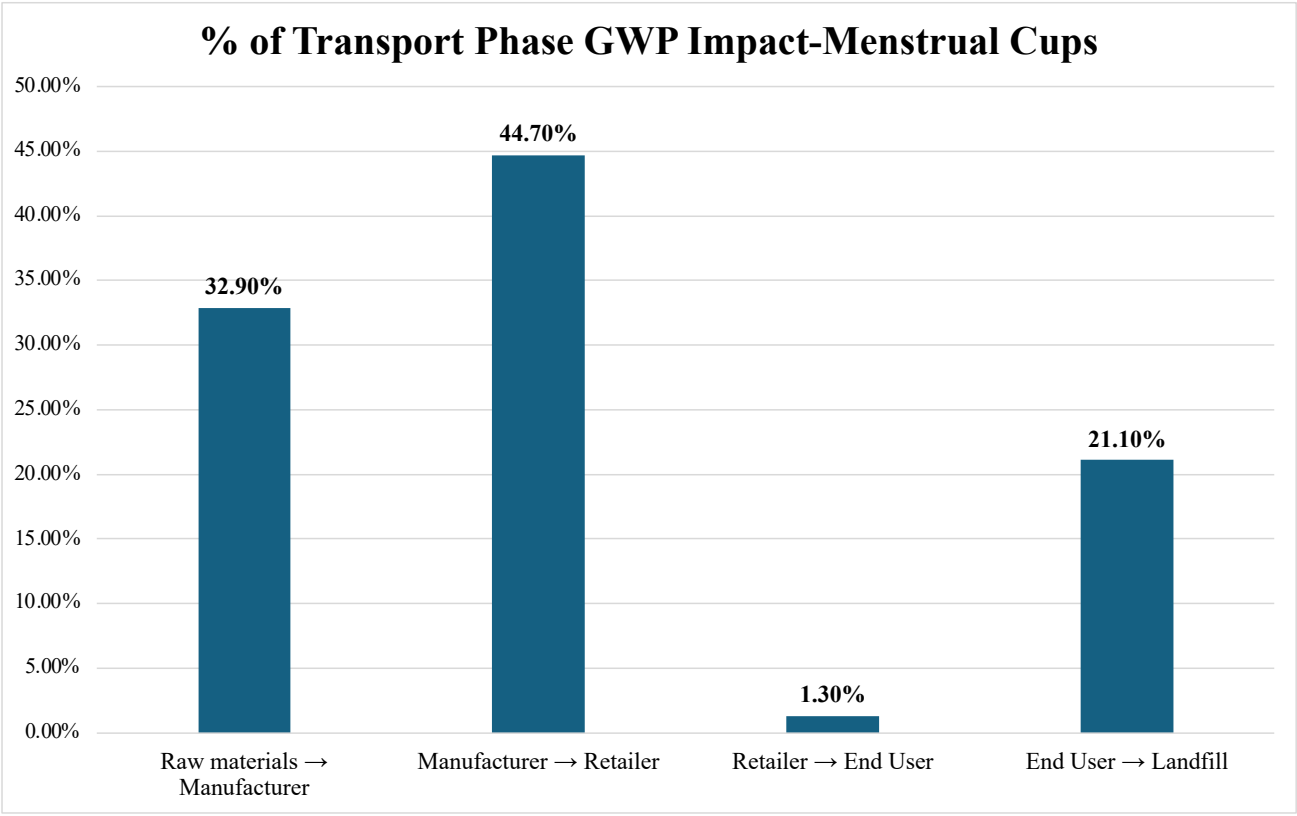


Figure 11:% of Transport Phase GWP – Menstrual Cup

2.4.4 Human Toxicity Potential

Human Toxicity Potential (HTP) assesses the potential harm to human health from chemical emissions throughout a product's life cycle. It is commonly expressed in kg 1,4-dichlorobenzene equivalents (kg 1,4-DCB-eq) and accounts for both cancer and non-cancer effects, incorporating emissions to air, water, and soil across all life cycle phases (Fourcassier et al., 2022).

This study uses HTP data to compare the potential human health impacts of 2400 single-use tampons with plastic applicators and one reusable silicone menstrual cup over a ten-year functional unit.

HTP Results and Comparison

Published LCAs consistently show that single-use tampons with plastic applicators exhibit significantly higher HTP than reusable menstrual cups. On average, the annual human toxicity burden for tampons has been estimated at ***~10^{-2} to 10^{-3} kg 1,4-DCB-eq***, while menstrual cups contribute only ***~10^{-4} to 10^{-5} kg 1,4-DCB-eq*** per year (Fourcassier et al., 2022). This represents a 10- to 100-fold reduction in human toxicity when switching to a menstrual cup.

The HTP associated with tampons primarily arises during the material production and manufacturing stages. These include emissions from rayon and cotton cultivation, bleaching, and the petrochemical processing of plastic applicators. For instance, bleaching of cellulose fibres can generate trace dioxins—highly toxic substances linked to carcinogenic and endocrine-disrupting effects (Fourcassier et al., 2022). Even small design differences, such as the inclusion of a plastic applicator, have been shown to increase HTP (Hait and Powers, 2019).

The menstrual cup's impacts however are concentrated in its manufacture from medical-grade silicone, a process involving relatively inert and non-toxic materials. Its use-phase contributions such as boiling for sterilisation are minor and do not contribute significantly to HTP (GreenDelta, 2020). End-of-life disposal impacts are also minimal as the cup is landfilled and lacks persistent leachates or incineration residues.

Furthermore, several studies using ReCiPe and USEtox frameworks have confirmed that menstrual cups contribute less than **1.5%** of the total human toxicity associated with disposables. Overall, the reusable menstrual cup emerges as the safer option for human health, both in magnitude and consistency across lifecycle phases (Fourcassier et al., 2022). *Table 9* compares tampons and menstrual cups across five life cycle stages for human toxicity potential, using a qualitative scale.

Table 9: Comparative Human Toxicity Potential (HTP) of Tampons and Menstrual Cups by Life Cycle Stage – Fourcassier et al. (2022); Hait and Powers (2019); GreenDelta (2020).

Life Cycle Stage	Tampons	Menstrual Cups
Material Production	**High** — fibre bleaching and plastics	**Moderate** — silicone polymer synthesis
Manufacture	**High** — energy-intensive + plastic forming	**Low** — one-time silicone moulding
Transport	**Moderate** — multi-stage logistics	**Low** — single product, lower volume
Use Phase	**Negligible**	**Very Low** — occasional boiling, no additives
End-of-Life Disposal	**Moderate** — incineration/landfill of plastics	**Very Low** — inert landfill, no leachate
Overall Toxicity Impact	**High**	**Very Low**

2.5 Interpretation

2.5.1 Lifecycle Hotspots and Emissions

For single-use tampons, the raw material extraction and processing phase was the dominant contributor to environmental impact, accounting for approximately **68%** of total CO_2 emissions. This is largely due to the energy-intensive production of rayon fibres, plastic applicators, and the use of chemical bleaching agents, all of which are known contributors to both climate change and human health risks (Hait and Powers, 2019; Fourcassier et al., 2022).

On the other hand, for the reusable menstrual cup, the use phase was the primary contributor to GWP, comprising over **61%** of total emissions. This was driven by monthly boiling for sterilisation, using an electric kettle rated at 2.5 kW for 5 minutes per cycle. Despite this, the cup's absolute GWP remains negligible relative to tampons.

Transport emissions were moderate for both products, but more significant for tampons **(≈11% of GWP)** due to the cumulative impact of shipping and local distribution of 2400 items over 10 years.

For cups, transport only contributed **2.2%** due to their one-off purchase and lower bulk weight.

End-of-life emissions were minimal in both base models (landfill assumed for both). However, real-world impacts could be more severe for tampons, which are often flushed — an action that contributes to sewer blockages and marine plastic pollution (Cabrera and Garcia, 2019).

Largest Emissions and Key Sources

- **Tampons**:
  - *GWP + HTP hotspots:* Rayon/cotton fibre production, polyethylene applicators.
- **Menstrual Cup**:
  - *GWP hotspot:* Monthly boiling using household energy.
  - *HTP impact:* Low — medical-grade silicone is chemically inert, non-leaching, and contributes minimally to long-term human toxicity (Fourcassier et al., 2022).

Comparison of Products

The reusable menstrual cup significantly outperformed tampons across both environmental indicators:

- **GWP**:
  - **Tampons**: 46.91 kg CO_2-eq
  - **Cup**: 0.34 kg CO_2-eq (**>99% reduction** over the 10-year functional unit)
- **HTP**:
  - Tampons exhibit up to **100 times greater toxicity potential** due to the cumulative impact of **bleaching agents, dioxins, and plastic-related emissions (**Hait and Powers, 2019).
  - Cups exhibit **negligible HTP** due to low-reactivity silicone and minimal chemical processing (Fourcassier et al., 2022; GreenDelta, 2020).

These findings align with LCA literature that consistently ranks cups as the most environmentally sustainable option among menstrual products (UNEP, 2022; Hait and Powers, 2019).

2.5.4 Appropriateness of Product Switch

A switch from single-use tampons to reusable cups is clearly encourageable from an environmental and health perspective with benefits including >99% GWP reduction and near-zero toxicity impacts.

While barriers to adoption such as public stigma, misinformation, or lack of sanitation infrastructure exist, these do not undermine the fundamental sustainability advantages of the cup (UNEP, 2022).

2.5.5 Strategies to Reduce Environmental Impacts

- **For Tampons**:
    - **Material reformulation**: Replace rayon with organic cotton and eliminate plastic applicators.
    - **End-of-life reform**: Promote non-flush labelling
- **For Menstrual Cups**:
    - **Use-phase optimisation**: Encourage low-energy sterilisation options, e.g. steam bags or solar kettles.
    - **Recycling innovation**: Develop silicone recovery or closed-loop take-back schemes.

Broader solutions may include policy intervention and public health education campaigns targeting menstrual stigma and access (UNEP, 2022).

3 ECONOMIC ASSESSMENTS

3.1 Data Collection and Assumptions

Cost data for both products were collected using UK-specific retail prices from Boots and Superdrug. The tampon (with plastic applicator) cost was based on a market-average 18-pack priced at **£3.50**, representative of leading brands like Tampax. This equates to **£465.50** over the ten-year functional unit. For the menstrual cup, a one-time purchase of a medical-grade silicone cup was modelled at **£20**, consistent with UK listings for popular brands like Mooncup (Statista, 2021).

Cost Breakdown and Key Contributors

Table 10: Cost Breakdown and Key Contributors Over 10 Years

Product	Unit Cost (£)	Quantity (10 years)	Total Cost (£)	Comparison of Major Cost Contributors
Tampons	£3.50 per pack of 18 = **£0.194**/tampon (Boots, 2020)	2400 tampons (134 packs)	**£465.50**	• Repetitive purchasing • Brand premiums on well-known products (Fourcassier et al., 2022) • Packaging & applicators increase cost but add no reuse value (Cohen et al., 2024)
Menstrual Cup	One-time purchase: **£20.00**	1 cup	**£20.00**	• Initial upfront purchase • No recurring material cost • Negligible sterilisation cost (~1.17 MJ/year ≈ <£1 over 10 years) (GreenDelta, 2020)

3.2 Cost-Per-Use Calculation

Cost-per-use was selected as an economic indicator to compare the economic sustainability of both products. This metric reflects how much a user pays each time they rely on either product for effective menstrual protection, based on market prices and typical use patterns in the UK.

- **Tampons:** A standard tampon provides up to 8 hours of protection, requiring approximately **4 tampons per day**. Assuming an average 5-day menstrual cycle per month, this totals 240 tampons per year, or 2400 over ten years (Hait & Powers, 2019).

  Total cost over 10 years = **£465.50**

  Cost per use = **£0.19**

- **Menstrual Cup:** Menstrual cups typically last up to ten years and are emptied twice daily during menstruation (Hait & Powers, 2019). This equates to **2 uses per day** for 5 days each month, over 12 months and 10 years = 1200 uses.

  Total cost = **£20.00**

  Cost per use = **£0.017**

3.3 Additional Estimation Method: Discounting and Time-Value

Although the primary analysis used cost per use, it's helpful to consider whether time-based economic evaluation such as discounting or Net Present Value (NPV) would meaningfully alter the findings.

Net Present Value (NPV) Approach

The menstrual cup involves a one-time cost at year zero, while tampons incur costs continuously over the ten-year period. Using a standard UK social discount rate of 3.5% (Hurst M, 2019), the discounted cumulative cost of tampons remains significantly higher than the menstrual cup even after accounting for the time value of money.

- **Tampons:** Annual cost = £46.50 Discounted over 10 years at 3.5%:

$$NPV = \sum_{t=1}^{10} \frac{£46.50}{1 + (0.035)^t} \approx £396.12$$

- **Menstrual Cup:** One-time cost = £20.00 Occurs at year 0, no discounting required.

3.4 Interpretation

The economic analysis clearly demonstrates that reusable menstrual cups are substantially more cost-effective than single-use tampons over a ten-year period, regardless of the estimation method applied.

- Cost per use was **£0.19** for tampons compared to only **£0.017** for the menstrual cup. This is more than **90%** cost reduction over the defined functional unit.
- The total cost of tampons reached **£465.50**, whereas the cup incurred a one-time cost of £20.00.
- Even when accounting for the time value of money using NPV calculations at a 3.5% discount rate, tampons still cost nearly **20 times more** than the menstrual cup over ten years.

These results reflect how product lifespan and reusability drive economic sustainability. Disposable products like tampons incur cumulative costs with every use, while the menstrual cup's single investment delivers repeated value. This difference is especially significant for low-income consumers, those facing menstrual poverty, or national public health initiatives seeking to improve access to affordable period care (Cohen et al., 2024), making menstrual cups a more sustainable choice from an economic perspective.

4 SOCIAL SUSTAINABILITY ASSESSMENT

Indicator: Consumer Health and Safety Risks

Methodology: Social Life Cycle Assessment (SLCA)

4.1 Goal and Scope of SLCA

This social assessment investigated the potential health and safety risks posed to consumers by two menstrual hygiene products over a 10-year usage period in the UK: single-use tampons with plastic applicators and reusable menstrual cups. The analysis followed the Social Life Cycle Assessment (SLCA) methodology and focused on the **end user stakeholder group**, assessing how product design, materials, and instructions for use could impact user well-being and safety.

The scope covered the use and end-of-life phases, where most consumer risk occurs. Key factors considered include toxic shock syndrome (TSS), chemical exposure, microbiological safety, and consumer misuse potential. The menstrual cup was assessed for risks related to insertion/removal, sterilisation, and allergic reactions. The tampon is assessed for TSS risk, residual chemical exposure (e.g. dioxins), and environmental contamination linked to improper disposal.

4.2 Inventory Analysis

Data were gathered from peer-reviewed clinical studies, medical reviews, systematic reviews on user safety, and manufacturer data. *Table 11* summarises known risks and their prevalence based on UK and international sources:

Table 11:Risks associated with tampon and menstrual cup use

Product	Risk Type	Notes / Evidence
Tampons	TSS risk	<1/100,000 users per year; linked to retained tampons and high absorbency use (CDC, 2022)
	Dioxin/chemical exposure	Very low levels, but chronic exposure risk debated (Cohen et al., 2024)
	Micro-tears / irritation	Friction may cause epithelial disruption (Fourcassier et al., 2022)
Menstrual cup	Incorrect insertion/removal	Reported pain, but <2% of users discontinue due to this (Cohen et al., 2024)
	Boiling/cleaning misuse	Risk of infection if improperly cleaned (Cohen et al., 2024)
	Silicone allergy (rare)	Medical-grade silicone is biocompatible (Cohen et al., 2024)

4.3 Impact Assessment

The impact assessment used a qualitative rating matrix *(Table 12)* based on SLCA guidance to categorise risk levels as High, Medium, or Low for each product across safety themes:

Table 12:Qualitative risk scoring for both products

Safety Risk Category	Tampons	Menstrual Cup
TSS / infection risk	High	Low
Chemical exposure	Medium	Low
Disposal-related health hazard	Medium	Low
Insertion/removal risk	Low	Medium
Cleaning/maintenance risk	N/A	Medium

These findings suggest that while both products are generally safe when used correctly, tampons present more severe but rare risks (e.g. TSS), whereas cups depend more heavily on user compliance with cleaning protocols (Cohen et al., 2024).

4.4 Interpretation

Tampons carry higher health risks due to TSS potential, micro-tears, and possible long-term chemical exposure from bleaching and fragrances. Although these are infrequent, their severity is high. However, menstrual cups pose moderate safety risks mainly linked to hygiene behaviour. Cups eliminate chemical exposure and reduce infection risks when used hygienically.

UK data suggests most users receive limited education on proper menstrual cup cleaning, highlighting a need for better instruction, which could reduce risk perception and improve adoption (Cohen et al., 2024).

5 INTERPRETATION & DESIGN RECOMMENDATIONS

5.1 Sensitivity Analysis

Assessment of Data Quality and Assumptions:

The inventory data sources selected for this assessment were generally highly reliable and transparent. *Table 13* assesses the reliability of the sources used in this study.

Table 13: Quality of Data Sources and Assumptions

Source Type	Sources Used	Pros	Cons
Research Papers	• Hait & Powers, 2019 • Fourcassier et al., 2022 • Hochwalt et al., 2023	• Peer-reviewed • Reliable • Specific	• Potentially too specific • Risk being outdated.
Government Reports	• UK Parliament, 2023 • DEFRA Waste Guidelines, 2021 • European Commission, 2020	• Reliable • Country-specific • Regulatory-backed	• Sometimes too broad
Large Organisation Databases	• GreenDelta • Granta EduPack Eco Audit Tool	• Widely accepted • User-friendly dataset • High transparency	• May lack specificity • May be unreliable
Private Reports and Databases	• Future Market Insights, 2025 • Boots Retail Database, 2024	• Frequently updated • Relevant industry-specific insights.	• May have limited transparency
Sector-specific Company Reports	• Tampax, 2025 • Diva International, 2015	• Highly specificity	• May be biased or lack transparency

A sensitivity analysis was conducted to evaluate the reliability of the sustainability outcomes of this study against inventory sources and assumptions used:

- **Tampon Disposal Method (Landfill vs. Flushing)**: While this study assumed 100% landfill disposal, evidence suggests a significant percentage of tampons are flushed down the toilet. Modelling a scenario where 30% of tampons are flushed increased the Global Warming Potential (GWP) slightly **(~4%)** due to added emissions from wastewater treatment and increased Human Toxicity Potential (HTP) due to marine plastic pollution and wastewater contamination (UNEP, 2022; UK Parliament, 2023). This further supports the switch from tampons to cups.

- **Energy Source for Cup Sterilisation**: Changing the energy source used for sterilising the menstrual cup from the standard UK electricity grid to renewable solar-heated boiling water substantially decreased the cup's GWP as it reduced emissions from the cup's use-phase by approximately **60%** (GreenDelta, 2020). This highlights the importance of renewable energy sources in further improving the cup's environmental advantage.

- **Menstrual Cup Lifespan**: This study assumed a 10-year lifespan for the menstrual cup. However, some guidelines recommend a conservative lifespan of 5 years. Under this scenario, a user would require two cups over ten years, doubling the total cost **to £40.00** and slightly increasing the GWP. Despite this, the menstrual cup remains significantly more cost-effective and environmentally friendly compared to tampons (GreenDelta, 2020).

These sensitivity analyses demonstrate that even under varied assumptions, the menstrual cup remains a more sustainable choice across all evaluated dimensions.

5.2 Comparative Evaluation & MCDA

A Multi-Criteria Decision Analysis (MCDA) was conducted to compare the menstrual cup and tampons across all four sustainability indicators: Global Warming Potential (GWP), Human Toxicity Potential (HTP), Cost per Use, and Consumer Health & Safety Risks.

Indicators were normalised on a scale from 0 (worst performance) to 1 (best performance) and then weighted according to their relative importance *(Table 14).* Given the UK's policy focus on achieving and maintaining minimal emissions, (Hurst M, 2019), GWP was given double the weighting (40%), while other indicators were weighted equally (20% each).

Table 14:MCDA Weighted Scoring Matrix

Sustainability Indicator	**Weighting**	**Tampons Score**	**Menstrual Cup Score**
Global Warming Potential (GWP)	40%	0.0	1.0
Human Toxicity Potential (HTP)	20%	0.1	0.9
Cost per Use	20%	0.1	0.9
Consumer Health & Safety Risks	20%	0.2	0.8
Total Weighted Score	**100%**	**0.08**	**0.92**

This analysis clearly indicates the menstrual cup as the superior option, significantly outperforming tampons across every indicator evaluated.

Trade-Offs

- **Tampons**: Easier initial adoption, convenience, but associated with higher environmental harm, significant ongoing costs, and greater consumer health risks.
- **Menstrual Cups**: Higher upfront cost, behavioural adaptation required, but significantly lower environmental and economic burdens.

Given these findings, transitioning from tampons to menstrual cups would be highly appropriate for achieving better sustainability outcomes, particularly in alignment with UK climate policy goals.

6.3 Design & Policy Recommendations

Based on the comparative MCDA outcomes, several recommendations emerge to improve sustainability and reduce the environmental, economic, and social impacts of menstrual products in the UK.

6.3.1 Product Design Improvements

Tampons:

- **Biodegradable & organic alternatives:** Transition to fully biodegradable applicators made from plant-based materials or certified organic cotton, reducing both environmental impact and toxicity potential (UNEP, 2022).
- **Reduced packaging and chemical treatments:** Eliminate unnecessary plastic wrappers, fragrances, and chemical bleaching agents to lower human health risks and environmental toxicity (Weir, 2015).

Menstrual Cups:

- **Material use and recycling:** Explore cup designs that minimise silicone use without compromising durability. Develop industry-standard closed-loop recycling systems for medical-grade silicone (Hochwalt et al., 2023).
- **Improved sterilisation solutions:** Promote low-energy sterilisation methods such as microwaveable steam bags or UV-based sterilisation boxes to further reduce lifecycle GWP (GreenDelta, 2020).

6.3.2 Policy and Regulatory Recommendations

Plastic Regulation & Extended Producer Responsibility (EPR):

- Implement regulations to phase out single-use plastic in menstrual products.
- Introduce EPR schemes for menstrual product manufacturers, mandating responsibility for disposal and recycling, thereby incentivising sustainable design.

Consumer Labelling and Education:

- Mandate clear labelling that explicitly discourages flushing of menstrual products, reducing marine pollution and associated toxicity.
- Establish government-backed consumer education campaigns highlighting the financial, health, and environmental benefits of menstrual cups.

Economic and Social Accessibility Measures

- Offer financial incentives or subsidies to reduce the upfront cost of menstrual cups, especially targeting low-income and vulnerable communities, to mitigate menstrual poverty and increase accessibility.
- Provide menstrual cups as part of government-funded menstrual hygiene schemes in schools, workplaces, and public institutions, promoting wider adoption and normalisation of reusable products.

7 CONCLUSIONS

7.1 Summary of Findings

This study undertook a sustainability assessment of single-use tampons with plastic applicators versus reusable silicone menstrual cups, comparing their performance over a ten-year functional unit defined as "effective menstrual protection for one individual." Four sustainability indicators were assessed: Global Warming Potential (GWP), Human Toxicity Potential (HTP), Cost per Use, and Consumer Health and Safety Risks, using LCA methodologies.

7.2 Key Conclusions

Tampons have significantly higher environmental impacts across all lifecycle stages mainly due to raw material extraction (rayon and plastic applicators), manufacturing emissions, and transportation. Menstrual cups demonstrated over **99%** reduction in GWP than tampons, proving their environmental sustainability over tampons. Cups also showed less human toxicity potential due to inert silicone than tampons which exhibited higher toxicity due to chemical processing and potential dioxin release.

The menstrual cup was much more cost-effective, with a cost per use of £0.017 compared to £0.19 for tampons—a reduction exceeding 90% over the ten-year analysis period. Even accounting for discounting (NPV), menstrual cups remained more economical than tampons.

Consumer health and safety assessments revealed tampons pose higher severity risks such as Toxic Shock Syndrome (TSS), chemical exposure, and irritation. Menstrual cups however had moderate risks mainly related to user compliance with hygiene practices.

Variations in assumptions, such as tampon disposal methods, energy sources for sterilisation, and menstrual cup lifespan, did not change the outcome of the study which suggests menstrual cups are the more sustainable alternative across all indicators.

7.3 Recommendations for Product Improvement

- **Tampons**:
  - Transition from plastic applicators to biodegradable, plant-based alternatives.
  - Eliminate chemical bleaching and fragrances to minimise human toxicity.
  - Improve packaging designs to reduce plastic content and improve recyclability
- **Menstrual Cups**:
  - Develop closed-loop silicone recycling schemes to further reduce lifecycle impacts.
  - Introduce low-energy sterilisation alternatives such as UV sterilisation or microwaveable steam bags.
  - Innovate cup design to use less silicone without compromising functionality

7.4 Recommendations for Further Work

Further work could include a detailed assessment of how personal and socioeconomic factors affect people's decisions around adopting menstrual cups. It would also be helpful to explore practical solutions and possible challenges for implementing silicone recycling programs across the UK. Additionally, broadening the scope of future assessments to include further lifecycle impact categories such as eutrophication, resource depletion, and broader ecological toxicity would provide a more holistic sustainability assessment.

REFERENCE LIST

Bogner, J., Pipatti, R., Hashimoto, S., Diaz, C., Mareckova, K., Diaz, L., Kjeldsen, P., Monni, S., Faaij, A., Qingxian Gao, Tianzhu Zhang, Mohammed Abdelrafie Ahmed, Sutamihardja, R.T.M. and Gregory, R. (2008). Mitigation of global greenhouse gas emissions from waste: conclusions and strategies from the Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report. Working Group III (Mitigation). *Waste Management & Research*, 26(1), pp.11–32. doi:https://doi.org/10.1177/0734242x07088433.

Boots (2020). *Beauty | Health | Pharmacy and Prescriptions - Boots*. [online] Boots.com. Available at: https://www.boots.com.

Cabrera, A. and Garcia, R. (2019). *The environmental & economic costs of single-use menstrual products, baby nappies & wet wipes: investigating the impact of these single-use items across Europe*. *ZWE, Wien*.

Cohen, E.S., Kouwenberg, L.H., Moody, K.S., Sperna Weiland, N.H., Kringos, D.S., Timmermans, A. and Hehenkamp, W.J., 2024. Environmental sustainability in obstetrics and gynaecology: a systematic review. *BJOG: An International Journal of Obstetrics & Gynaecology*, *131*(5), pp.555-567.

Contou, D., Colin, G., Travert, B., Jochmans, S., Conrad, M., Lascarrou, J.B., Painvin, B., Ferré, A., Schnell, D., La Combe, B. and Coudroy, R., 2022. Menstrual toxic shock syndrome: a French nationwide multicenter retrospective study. *Clinical Infectious Diseases*, *74*(2), pp.246-253.

Davidson, A. (2012). Narratives of Menstrual Product Consumption. *Bulletin of Science, Technology & Society*, 32(1), pp.56–70. doi:https://doi.org/10.1177/0270467612444579.

DEFRA (2021). *Department for Environment, Food & Rural Affairs*. [online] GOV.UK. Available at: https://www.gov.uk/government/organisations/department-for-environment-food-rural-affairs.

DIVA Canada. (2025). *DIVA™ Cup Model 0*. [online] Available at: https://shopdiva.ca/products/diva-cup-model-0 [Accessed 26 May 2025].

Durante, S., Comoglio, M., Ridgway, N. and Qin, Y. (2015). Chapter 34Life Cycle Assessment in Nanotechnology, Materials and Manufacturing. In: *Micro and Nano Technologies*. [online] Boston: William Andrew Publishing, pp.775–804. doi:https://doi.org/10.1016/B9780323311496.000347.

Einhorn Product GmbH (2020) (2022). *Comparative Life Cycle Assessment of Menstrual Products – Final Report. GreenDelta* . [online] Google.com. Available at: https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://einhorn.my/wp-content/uploads/2022/03/Comparative-Life-Cycle-Assessment-of-Menstrual-Products.pdf&ved=2ahUKEwjGh-T28rCNAxVmUUEAHcqFLX8QFnoECBgQAQ&usg=AOvVaw0vYfEFtFUv21idVhq9dS7Y [Accessed 20 May 2025].

Faraca, G., PEREZ, C.M., Alfonso, L.B., PEREZ, A.Z., Kowalska, M.A. and Wolf, O. (2023). *Revision of EU ecolabel criteria for absorbent hygiene products and reusable menstrual cups (previously absorbent hygiene products)*.

Fourcassier, S., Douziech, M., Pérez-López, P. and Schiebinger, L. (2022). Menstrual products: A comparable life cycle assessment. *Cleaner Environmental Systems*, 7, p.100096.

Friends of the Earth (2018). *Plastic periods: menstrual products and plastic pollution*. [online] Friends of the Earth. Available at: https://friendsoftheearth.uk/sustainable-living/plastic-periods-menstrual-products-and-plastic-pollution.

Hait, A. and Powers, S.E. (2019). The Value of Reusable Feminine Hygiene Products Evaluated by Comparative Environmental Life Cycle Assessment. *Resources, Conservation and Recycling*, 150(1), p.104422. doi:https://doi.org/10.1016/j.resconrec.2019.104422.

Hochwalt, A.E., Abbinante-Nissen, J.M., Bohman, L., Hattersley, A.M., Hu, P., Streicher-Scott, J.L., Teufel, A.G. and Woeller, K.E. (2023). The safety assessment of tampons: illustration of a comprehensive approach for four different products. *Frontiers in reproductive health*, [online] 5. doi:https://doi.org/10.3389/frph.2023.1167868.

Hurst, M., 2019. The Green Book: Central Government Guidance on Appraisal and Evaluation: HM Treasury, London, UK: OGL Press, 2018 (also available online on www. gov. uk/government/publications), ISBN 978-1-912225-57-6.

Lovell, H., Boston, C., Hardiman, M., Pepin, N. and Rumble, H., 2023. Written evidence to the UK Parliament Environmental Audit Sub-Committee on Polar Research Special Inquiry on'The UK and the Arctic Environment'.

Ramsay, C., Hennegan, J., Douglass, C.H., Eddy, S., Head, A. and Lim, M.S.C. (2023). Reusable period products: use and perceptions among young people in Victoria, Australia. *BMC Women's Health*, 23(1). doi:https://doi.org/10.1186/s12905-023-02197-3.

Shout Out UK. (2024). *How Do Period Products Affect the Environment?* [online] Available at: https://www.shoutoutuk.org/2024/06/05/how-do-period-products-affect-the-environment/.

Statista. (n.d.). *Leading brands of tampons in the UK 2018-2020*. [online] Available at: https://www.statista.com/statistics/305057/leading-brands-of-tampons-in-the-uk/.

Tampax.co.uk. (2021). *Quality and composition of Tampax tampons*. [online] Available at: https://tampax.co.uk/en-gb/safety-promise/what-are-tampax-tampons-made-of/ [Accessed 26 May 2025].

UNEP, U. (2021). *Single-use menstrual products and their alternatives: Recommendations from life cycle assessments*. *Nairobi: United Nations Environment Programme*.

UNICEF (2020). *Menstrual hygiene*. [online] UNICEF. Available at: https://www.unicef.org/wash/menstrual-hygiene.

Vilabrille Paz, C., Ciroth, A., Mitra, A., Birnbach, M. and Wunsch, N. (2020)Comparative Life cycle assessment of menstrual products. GreenDelta GmbH, commissioned by einhornproducts GmbH https://www.greendelta.com/project/lcaofmenstrualproducts/

Weir, C.S., 2015. In The Red: A private economic cost and qualitative analysis of environmental and health implications for five menstrual products. *Dalhousie University*, p.6.

www.futuremarketinsights.com. (n.d.). *Feminine Hygiene Products Market*. [online] Available at: https://www.futuremarketinsights.com/reports/feminine-hygiene-products-market.